

ENTERPRISE CYBER SECURITY GOVERNANCE FRAMEWORK FOR A FINANCIAL INSTITUTION

Project Report

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ABSTRACT

This project presents an Enterprise Cyber Security Governance Framework for a financial institution. It focuses on identifying information assets, assessing cybersecurity risks, defining security policies, and recommending governance practices to improve organizational security.

The project follows industry best practices and highlights the importance of information security governance, risk management, compliance, and employee awareness in protecting sensitive financial information.

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1. INTRODUCTION

Cybersecurity governance is the process of managing and protecting an organization's information assets through effective policies, risk management, and security controls. Financial institutions are prime targets for cyber attacks because they handle sensitive customer and financial information.

A strong cybersecurity governance framework helps organizations reduce cyber risks, ensure regulatory compliance, and maintain customer trust. It establishes clear security policies, assigns responsibilities, and promotes a security-aware culture.

This project presents a simplified Enterprise Cyber Security Governance Framework for a financial institution. It focuses on identifying key information assets, assessing cyber risks, defining essential security policies, and recommending practical security measures to improve the organization's cybersecurity posture.

Objectives

- Identify critical information assets.
- Assess cybersecurity risks.
- Develop essential information security policies.
- Recommend governance and security controls.
- Improve organizational cyber resilience.

Scope

.This project focuses on cybersecurity governance for a financial institution.

.It covers asset identification, risk assessment, information security policies, and security recommendations.

.The project is intended for academic purposes and demonstrates cybersecurity governance concepts based on industry best practices.

2. ORGANIZATION OVERVIEW

XYZ Bank is a medium-sized financial institution that provides banking services, digital payments, internet banking, and mobile banking to its customers. The organization stores sensitive customer information, financial records, and transaction data. Due to the increasing number of cyber threats, the bank requires a strong cybersecurity governance framework to protect its critical assets and ensure regulatory compliance.

Critical Information Assets

- .Customer Database- Stores customer information
- .Internet Banking System- Online banking services
- .Core Banking Server- Financial transactions
- .Employee Laptops- Daily operations
- .Email System- Internal & external communication
- .Backup Server- Data recovery

Existing Security Controls

The organization currently uses:

- Firewall Protection
- Antivirus Software
- Secure Password Policy
- Data Backup
- Network Monitoring
- Employee Security Awareness Training

Asset Identification & Risk Assessment

Risk Assessment Table

.Customer Database- Data Breach

.Internet Banking- Phishing Attack

.Core Banking Server- Ransomware

.Email System- Phishing

.Employee Laptops- Malware

.Backup Server- Unauthorized Access

Risk Summary

The risk assessment identified the Customer Database, Core Banking Server, Internet Banking System, and Backup Server as high-risk assets. These systems require strong security controls, regular monitoring, and continuous risk assessment to ensure the confidentiality, integrity, and availability of financial information.

Information Security Policies

The following information security policies are recommended for the organization:

1. Password Policy

- Use strong passwords with at least 12 characters.
- Enable Multi-Factor Authentication (MFA).
- Change passwords every 90 days.

2. Access Control Policy

- Grant access based on job roles.
- Remove inactive user accounts.
- Review user access regularly.

3. Data Protection Policy

- Encrypt sensitive customer data.
- Perform regular backups.
- Restrict unauthorized data access.

4. Incident Response Policy

- Report security incidents immediately.
- Isolate affected systems.
- Preserve digital evidence.
- Document all response activities.

SECURITY RECOMMENDATIONS

- Enable Multi-Factor Authentication (MFA).
- Conduct regular cybersecurity awareness training.
- Update operating systems and applications regularly.
- Perform quarterly security risk assessments.
- Monitor network activity continuously.
- Encrypt sensitive financial information.
- Test backup and disaster recovery procedures.
- Conduct annual cybersecurity audits.

CONCLUSION

A strong cybersecurity governance framework is essential for protecting financial institutions against evolving cyber threats.

This project demonstrated how governance, risk assessment, security policies, and employee awareness contribute to a secure environment.

By implementing the recommended controls, organizations can improve their cybersecurity posture, reduce operational risks, and protect sensitive financial information.

REFERENCES

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